

HAT-P-36b

R-0945

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230308 · created 2026-07-29T13:38:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460011.7859 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.006 +/- 0.045
Transit depth (Rp/Rs)^2	0.0032 +/- 0.051 [%]
Orbital Inclination (inc)	84.23 +/- 2.82
Airmass coefficient 1 (a1)	0.066 +/- 0.036
Airmass coefficient 2 (a2)	2.13 +/- 0.49
Scatter in the residuals of the lightcurve fit is	45.77 %
Best Comparison Star	None
Optimal Aperture	7.77
Optimal Annulus	9.06
Transit Duration (day)	0.079 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.006 ± 0.045 vs 0.126 (deviation 2.67σ)
Duration fit vs published	0.079 d vs 0.0925 d (deviation 1.13σ)
Mid-time O–C	-11.3 min
Depth SNR	0.1
Residual scatter	45.77 %

Light curve

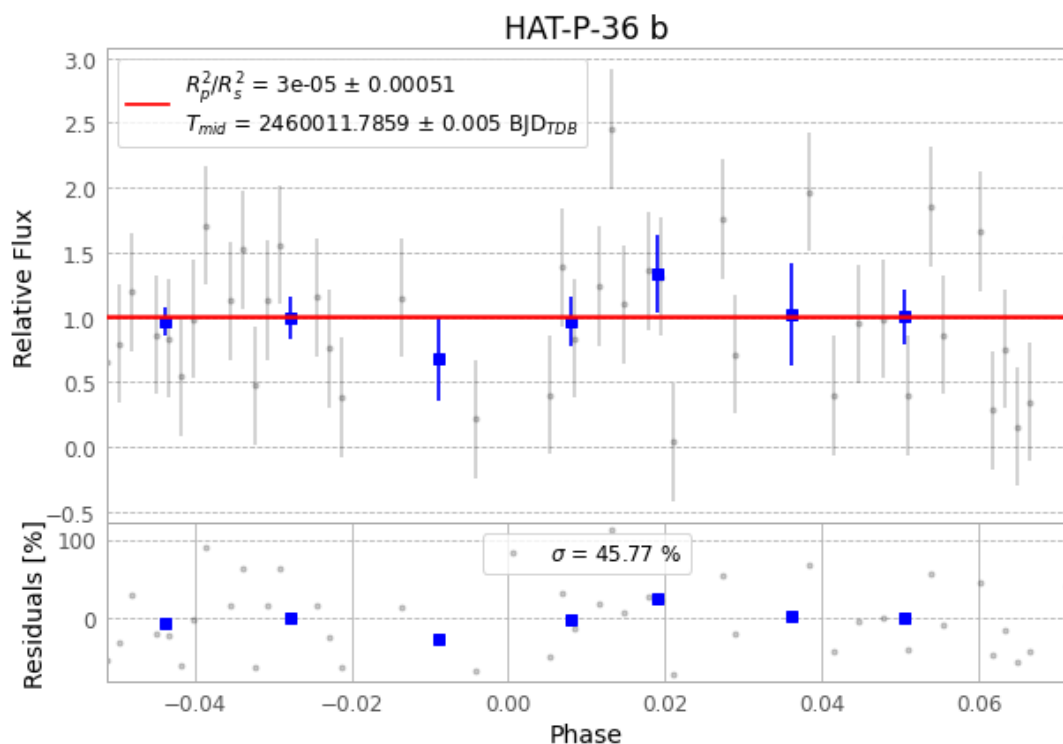


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-07.png (SHA-256 d26991135fd90afc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [538, 90], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 435b285e2c2161a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), HATP-36230308050315.FITS → HATP-36230308090015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230308.log (129d81da6565...), FinalLightCurve_HAT-P-36 b_2023-03-07.png (d26991135fd9...), FinalLightCurve_HAT-P-36 b_2023-03-07.csv (4e7fff4c088e...)

Compute resources

Wall time 380.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 13:38Z.